

Assignment - 6.1

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Batch_17

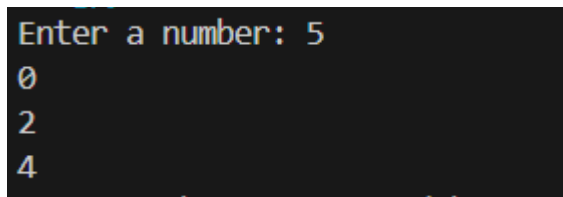
Task1 : Use an AI code completion tool to generate a loop-based program.

Prompt : # generate code to print even numbers from 0 to n using loop

Code :

```
n = int(input("Enter a number: "))  
for i in range(0, n + 1, 2):  
    print(i)
```

output :



```
Enter a number: 5  
0  
2  
4
```

TASK – 2: Use an AI code completion tool to combine loops and conditionals.

CODE :

```
numbers = list(map(int, input("Enter numbers separated by spaces:  
").split()))  
  
even_count = 0  
odd_count = 0  
  
for number in numbers:  
    if number % 2 == 0:
```

```
        even_count += 1
    else:
        odd_count += 1
print(f"Even numbers count: {even_count}")
print(f"Odd numbers count: {odd_count}")
```

OUTPUT :

```
Enter numbers separated by spaces: 5 4 5 4 3 2 2
Even numbers count: 4
Odd numbers count: 3
```

TASK – 3 : Use an AI tool to complete a Python class that validates user input.

PROMPT : #Generate a Python class User that validates age and email using CONDITIONAL STATEMENTS by user input

CODE : import re

class User:

def __init__(self, age, email):

self.age = age

self.email = email

self.validate_age()

self.validate_email()

def validate_age(self):

if self.age < 0 or self.age > 120:

raise ValueError("Age must be between 0 and 120.")

```

else:

    print("Age is valid.")

def validate_email(self):

    email_pattern = r'^[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}$'

    if not re.match(email_pattern, self.email):

        raise ValueError("Invalid email format.")

    else:

        print("Email is valid.")
age = int(input("Enter your age: "))
email = input("Enter your email: ")
user = User(age, email)

```

OUTPUT :

```

Enter your age: 34
Enter your email: nhfjhdsshjv232.com
Age is valid.

```

TASK- 4 : Use an AI code completion tool to generate a Python class for managing student details.

PROMPT : # Generate a Python class Student with attributes (name, roll number,marks) and methods to calculate total and average marks.

CODE :

```
class Student:

    def __init__(self, name, roll_number, marks):

        self.name = name

        self.roll_number = roll_number

        self.marks = marks


    def calculate_total(self):

        return sum(self.marks)


    def calculate_average(self):

        return sum(self.marks) / len(self.marks)

name = input("Enter student name: ")
roll_number = input("Enter roll number: ")
marks = list(map(int, input("Enter marks separated by spaces: ").split()))
student = Student(name, roll_number, marks)
total_marks = student.calculate_total()
average_marks = student.calculate_average()
print(f"Total Marks: {total_marks}")
print(f"Average Marks: {average_marks}")
```

OUTPUT :

```
Enter student name: vidj
Enter roll number: 33 44 55 66 77 66
Enter marks separated by spaces: 33 44 55 33 22
Total Marks: 187
Average Marks: 37.4
```

TASK-5 : Use an AI tool to generate a complete Python program using classes, loops, and conditionals together

PROMPT : #Generate a Python program for a simple bank account system using class, loops, and conditional statements.

CODE :

```
class BankAccount:

    def __init__(self, account_holder, balance=0):

        self.account_holder = account_holder

        self.balance = balance

    def deposit(self, amount):

        if amount > 0:

            self.balance += amount

            print(f"Deposited: {amount}. New balance: {self.balance}")

        else:

            print("Deposit amount must be positive.")

    def withdraw(self, amount):

        if amount > 0:
```

```

        if amount <= self.balance:

            self.balance -= amount

            print(f"Withdrew: {amount}. New balance: {self.balance}")

        else:

            print("Insufficient funds.")

    else:

        print("Withdrawal amount must be positive.")


def display_balance(self):

    print(f"Account Holder: {self.account_holder}, Balance: {self.balance}")

account_holder = input("Enter account holder name: ")

account = BankAccount(account_holder)

while True:

    print("\nOptions: 1. Deposit 2. Withdraw 3. Display Balance 4. Exit")

    choice = input("Choose an option (1-4): ")

    if choice == '1':

        amount = float(input("Enter amount to deposit: "))

        account.deposit(amount)

    elif choice == '2':

        amount = float(input("Enter amount to withdraw: "))

        account.withdraw(amount)

    elif choice == '3':

        account.display_balance()

    elif choice == '4':

        print("Exiting...")

```

```
break
```

```
else:
```

```
    print("Invalid choice. Please try again.")
```

OUTPUT ;

```
Options: 1. Deposit 2. Withdraw 3. Display Balance 4. Exit  
Choose an option (1-4): 2  
Enter amount to withdraw: 2333  
Insufficient funds.
```